

PUZZLEVQA: Diagnosing Multimodal Reasoning Challenges of Language Models with Abstract Visual Patterns

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🔗: <https://github.com/declare-lab/LLM-PuzzleTest>

🌐: <https://puzzlevqa.github.io/>

Abstract

Large multimodal models extend the impressive capabilities of large language models by integrating multimodal understanding abilities. However, it is not clear how they can emulate the general intelligence and reasoning ability of humans. As recognizing patterns and abstracting concepts are key to general intelligence, we introduce PUZZLEVQA, a collection of 2000 puzzle instances based on abstract patterns. With this dataset, we evaluate large multimodal models with abstract patterns based on fundamental concepts, including colors, numbers, sizes, and shapes. Through our experiments on state-of-the-art large multimodal models, we find that they are not able to generalize well to simple abstract patterns. Notably, GPT-4V achieves a score of 46.4% on single-concept puzzles, which shows that state-of-the-art models struggle on our dataset. To diagnose the reasoning challenges in large multimodal models, we progressively guide the models with our ground truth reasoning explanations for visual perception, inductive reasoning, and deductive reasoning. Our systematic analysis finds that the main bottlenecks of GPT-4V are weaker visual perception and inductive reasoning abilities. Through this work, we hope to shed light on the limitations of large multimodal models and how they can better emulate human cognitive processes in the future¹.

1 Introduction

Rapid advances in large language models have demonstrated remarkable capabilities across diverse language tasks and applications (Bubeck et al., 2023; Brown et al., 2020; Touvron et al., 2023b). To enable more general capabilities, large

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¹Our data and code are released at <https://github.com/declare-lab/LLM-PuzzleTest>.

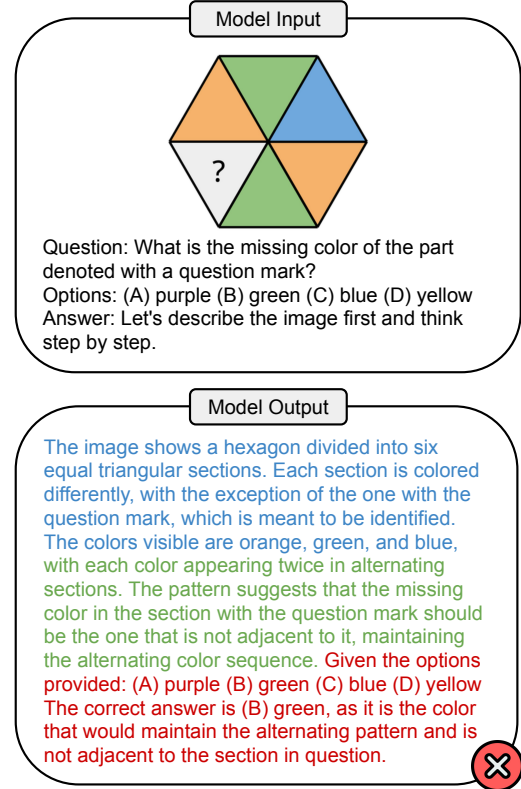


Figure 1: An example question which involves the color concept in PUZZLEVQA, and an incorrect answer from GPT-4V. There are generally three stages that can be observed in the solving process: visual perception (blue), inductive reasoning (green), and deductive reasoning (red). Here, the visual perception was incomplete, causing a mistake during deductive reasoning.

multimodal models were introduced by integrating large language models with multimodal understanding (Yue et al., 2023; Yang et al., 2023; OpenAI, 2023). However, it is not clear how large multimodal models can emulate the general intelligence and reasoning ability of humans (Qiu et al., 2024). Specifically, we aim to explore how large multimodal models can emulate cognitive processes to perceive and interpret information, extrapolate from observations to broader generaliza-

tions, and apply general principles to solve specific problems (Piaget, 1976). Furthermore, we are interested in understanding how well the models can reason about fundamental concepts such as numbers, colors, shapes, and size (Tong et al., 2024; Sharma et al., 2024).

As pattern recognition and abstracting concepts are at the heart of general intelligence (Tenenbaum, 2018; Carey, 2000; Cole, 1996), we believe that abstract patterns are a suitable testbed for evaluating reasoning ability in large multimodal models. As shown in Figure 1, abstract patterns enable us to focus on one or more abstract concepts, and decompose the multimodal reasoning process into several stages that mimic human cognitive processes (Piaget, 1976). Firstly, the model requires visual perception to understand and interpret the abstract image in the input. Secondly, the model requires inductive reasoning to relate the observations shown and form a hypothesis for the underlying pattern. Thirdly, the model requires deductive reasoning to apply the general principle of the pattern to solve the specific problem at hand. While the abstract patterns may seem simple, we surprisingly find that even advanced large multimodal models such as Gemini Pro (Gemini Team, 2023) and GPT-4V (OpenAI, 2023) struggle to understand them.

Puzzles are problems that require ingenuity and creativity to solve, and they can serve as valuable tools for cognitive development and assessment (Zhang et al., 2019). Hence, we propose the PUZZLEVQA dataset to systematically evaluate and diagnose the reasoning challenges in large multimodal models. Our dataset consists of diverse multimodal puzzles that focus on abstract patterns with fundamental concepts including numbers, colors, shapes, and size. We design and automatically construct the dataset through multimodal templates, enabling us to generate large numbers of puzzles without costly human annotation (Ding et al., 2022). To support interpretability and systematic investigation of reasoning challenges in multimodal models, we also construct the ground truth reasoning explanations for each puzzle. Compared to existing datasets for visual question answering, PUZZLEVQA focuses specifically on how large multimodal models can mimic general cognitive processes such as inductive and deductive reasoning. As we focus on how models can generalize to novel problems, similar to fluid intelligence in humans (Cattell, 1963), our dataset in the abstract domain poses challenges for existing models without re-

quiring extensive world knowledge.

Through our investigation of leading large multimodal models, we find that existing models are not able to generalize well to simple abstract patterns. Notably, GPT-4V achieves a score of 46.4% on single-concept puzzles, which shows that state-of-the-art models struggle on our dataset. Our analysis reveals that its main bottlenecks are weaker visual perception and inductive reasoning abilities. Hence, our main contributions include:

1. To investigate the cognitive and reasoning abilities of large multimodal models, we propose to leverage abstract patterns.
2. We introduce PUZZLEVQA, an automatically generated and diverse dataset of 2000 multimodal samples with reasoning explanations.
3. Our experiments show that even advanced large multimodal models do not generalize well to abstract patterns, and we show how to identify their reasoning bottlenecks.

2 Background: Cognitive Theories

To understand how large multimodal models can better mimic human thought processes and general intelligence, we first ground our study with relevant cognitive theories.

2.1 Fluid and Crystallized Intelligence

The Cattell-Horn theory (Cattell, 1963) of cognitive abilities distinguishes between two types of intelligence: fluid intelligence, which involves the ability to solve novel problems without relying on previously acquired knowledge, and crystallized intelligence, which involves the use of knowledge, skills, and experience. Fluid intelligence in humans could parallel large multimodal models’ ability to solve new, unseen problems through pattern recognition and problem-solving strategies. On the other hand, crystallized intelligence could be akin to how the models leverage accumulated world knowledge from training data to understand and interact with the world (Sumers et al., 2023). As many works have focused on how models can leverage specialized knowledge (Yue et al., 2023), we instead focus on how they may emulate fluid intelligence to solve novel problems through abstract patterns.

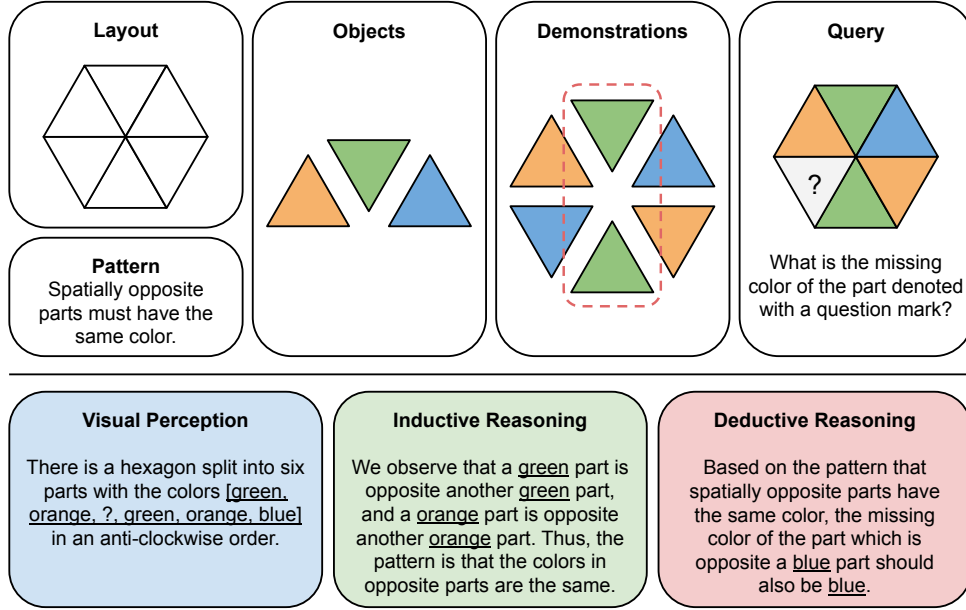


Figure 2: Illustration example of components (top) and reasoning explanations (bottom) for abstract puzzles in PUZZLEVQA. To construct each puzzle instance, we first define the layout and pattern of a multimodal template, and populate the template with suitable objects that demonstrate the underlying pattern. For interpretability, we also construct ground truth reasoning explanations to interpret the puzzle and explain the general solution stages.

2.2 Cognitive Development

Piaget’s Stages of Cognitive Development (Piaget, 1976) can provide a framework for progressing from basic sensory experiences to complex abstract reasoning and problem-solving. While we note that the large multimodal models do not develop in the same organic and experiential manner as humans, we are guided to explore how the models can emulate different stages of cognitive abilities. Concretely, through abstract patterns, we can evaluate how the models perceive multimodal information, reason inductively to extrapolate from observations to broader generalizations, and apply general principles to deduce the solution for specific problems.

Sensorimotor. This stage underpins visual perception, where individuals learn to coordinate sensory experiences through interactions with the environment. To emulate this cognitive stage, we would expect models to identify simple shapes or colors but lack higher-level reasoning. Hence, we set the foundation for later stages by exploring abstract patterns based on fundamental concepts including colors, numbers, shapes, and size.

Preoperational. At this stage, individuals develop symbolic thinking, which is crucial for understanding representations in visual contexts and beginning to engage in simple, inductive reasoning processes. Models that mimic this stage should be

able to perform basic reasoning about objects or concepts, but with limited understanding of abstract relationships or performing logical operations.

Concrete Operational. This stage is closely related to inductive reasoning, as individuals learn to think logically about concrete events and solve problems based on visible patterns and relationships. We would expect models that are analogous to this stage to have the ability to draw logical conclusions from specific instances and start to apply these conclusions to solve problems. Hence, we consider inductive reasoning as an integral part of understanding abstract patterns.

Formal Operational. This stage is essential for deductive reasoning and abstract thinking, allowing individuals to hypothesize and think about theoretical scenarios, which are skills necessary for solving complex problems. At this stage, we would expect comparable models to effectively induce general principles or hypotheses from observations and logically deduce specific outcomes, even in abstract or novel contexts. Thus, we consider deductive reasoning as critical to solving abstract problems.

3 PuzzleVQA Dataset

Despite the impressive capabilities of large multimodal models, we do not fully understand how they solve multimodal problems through reasoning.